

4(a)	correct formula ; (×) 1200 ; A 1171 1229 or 1228.6 (41 000 ÷ 35) (43 000 ÷ 35) <i>max 1 if units given</i>	2
4(b)(i)	<i>any two from</i> <u>DNA</u> , replication / AW ; A S phase I chromosome replication (early interphase) chromosome becomes two chromatids ; A described DNA checked for errors / errors in DNA repaired ; synthesis of, (growth) protein / enzymes (for growth) ; cell growth ; organelle synthesis ; centrioles replicate ; AVP ; e.g. <i>ref.</i> to checkpoint(s) nucleotide synthesis	2
4(b)(ii)	<i>any two from</i> <i>ref.</i> target cells are endothelial cells ; VEGF / cell signalling molecule, binds / AW, to receptor ; R receptor cells <i>ref.</i> specificity (of receptors) / VEGF / cell signalling molecule, complementary to receptors ; suggestion of detail following binding ; e.g. triggers secondary messenger activates enzymes / enzyme cascade signal transduction phosphorylation events	2

Question	Answer	Marks
4(c)	<i>any four from</i> VEGF protein / antigen, injected into, mouse / small mammal ; time for immune response to occur ; A description of event in immune response plasma cells / B-lymphocytes / splenocytes, extracted from mouse ; plasma cells / AW, fused with, myeloma / cancer, cells ; (to produce) hybridoma cells ; clone hybridoma cells ; separate cells and culture in individual wells ; screen for cells secreting desired antibody ; hybridoma cells grown in large scale culture / AW ; AVP ; e.g. <i>ref.</i> to fusogen	4
4(d)	<i>any one from</i> monoclonal antibodies identified, as foreign / non-self / act as an antigen ; (monoclonal antibodies) stimulate an, immune / allergic, response ; AVP ; e.g. (antibodies destroyed so) not enough antibody (to be effective)	1